Available online at www.sciencedirect.com
ScienceDirect
Neuropharmacology 51 (2006) 1109–1119
NEURO PHARMACOLOGY
www.elsevier.com/locate/neuropharm

Cholinergic receptor-independent dysfunction of mitochondrial respiratory chain enzymes, reduced mitochondrial transmembrane potential and ATP depletion underlie necrotic cell death induced by the organophosphate poison mevinphos

J.Y.H. Chan [a], S.H.H. Chan [b,c], K.Y. Dai [b], H.L. Cheng [b], J.L.J. Chou [b], A.Y.W. Chang [b,c,*]

[a] *Department of Medical Education and Research, Kaohsiung Veterans General Hospital, Kaohsiung 81346, Taiwan, ROC*
[b] *Center for Neuroscience, National Sun Yat-sen University, Kaohsiung 80424, Taiwan, ROC*
[c] *Center for Gene Regulation and Signal Transduction Research, National Cheng Kung University, Tainan 70101, Taiwan, ROC*

Received 12 May 2006; received in revised form 14 June 2006; accepted 23 June 2006

Abstract

Our current understanding of the nature of cell death that is associated with fatal organophosphate poisoning and the underlying cellular mechanisms is surprisingly limited. Taking advantage of the absence in an in vitro system of acetylcholinesterase, the pharmacological target of organophosphate compounds, the present study evaluated the hypothesis that the repertoire of cholinergic receptor-independent cellular events that underlie fatal organophosphate poisoning entails induction of mitochondrial dysfunction, followed by bioenergetic failure that leads to necrotic cell death because of ATP depletion. Pheochromocytoma PC12 cells incubated with the organophosphate pesticide mevinphos (0.4 or 4 μmol) for 1 or 3 h underwent a dose-related and time-dependent loss of cell viability that was not reversed by muscarinic (atropine) or nicotinic (mecamylamine) blockade. This was accompanied by depressed NADH cytochrome *c* reductase, succinate cytochrome *c* reductase or cytochrome *c* oxidase activity in the mitochondrial respiratory chain, reduced mitochondrial transmembrane potential, decreased ATP concentration, elevated ADP/ATP ratio, increased lactate dehydrogenase release and necrotic cell death. We conclude that Mev induces cholinergic receptor-independent necrotic cell death by depressing the activity of Complexes I to IV in the mitochondrial respiratory chain, eliciting reduction in mitochondrial transmembrane potential, depleting intracellular ATP contents and damaging cell membrane integrity.
© 2006 Elsevier Ltd. All rights reserved.

Keywords: Mevinphos; Mitochondrial respiratory chain enzymes; Mitochondrial permeability transition; Cell membrane integrity; PC12 cells

1. Introduction

The organophosphate compounds are among the most widely used pesticides in the world (Balali-Mood and Shariat, 1998) and represent one of the oldest agents used in chemical warfare (McCauley et al., 2001). Recent years saw renewed attentions to these compounds because of the impending threat of their deployment by militias or terrorists (Lallement et al., 1999), and the increased incidences of accidental or suicidal organophosphate poisoning because of easy access to pesticides in agriculture and home gardening (Bardin et al., 1994). The clinical presentations of organophosphate poisoning, including salivation, muscle fasciculation, stupor, cardiopulmonary collapse, respiratory failure, coma or death (Bardin and van Eeden, 1990), are generally believed to entail overstimulation of muscarinic receptors at peripheral and central synapses by the accumulated acetylcholine that results from inhibition of acetylcholinesterase (Tafuri and Roberts, 1987). Thus, a time-honored treatment scheme still in practice calls

* Corresponding author. Center for Neuroscience, National Sun Yat-sen University, Kaohsiung 80424, Taiwan, ROC. Tel.: +886 7 525 3629; fax: +886 7 525 5801.
E-mail address: achang@mail.nsysu.edu.tw (A.Y.W. Chang).

0028-3908/$ - see front matter © 2006 Elsevier Ltd. All rights reserved.
doi:10.1016/j.neuropharm.2006.06.024

for administration of atropine immediately after the arrival of patients at medical facilities (Bardin and van Eeden, 1990; Yen et al., 2000). Of note is that despite repeated dosing of atropine, some of our patients still succumbed to acute organophosphate poisoning (Yen et al., 2000). Given that brain stem death is the legal definition of death in the UK (Conference of Medical Royal Colleges and their Faculties in the United Kingdom, 1976, 1979), United States (Anonymous, 1981) or Taiwan (Hung and Chen, 1995), this clinical observation suggests that cellular mechanisms that are independent of cholinergic neurotransmission in the brain stem may underlie fatal organophosphate poisoning.

As the primary organelle for cellular ATP production (Hatefi, 1985), the mitochondrion presents itself as an important target for fatal cellular insults. In this context, two notions on the nature of cell death and its relationship with mitochondrial dysfunction and bioenergetic failure are of interest. One stipulates that the mitochondrial permeability transition is involved in both apoptotic and necrotic cell death (Ankarcrona et al., 1995; Kroemer, 1999; Crompton, 2000; Uchiyama et al., 2002; Kim et al., 2003). The other advocates that a critical determinant of the eventual cell death fate resides in intracellular ATP concentration (Eguchi et al., 1997; Leist et al., 1997; Kroemer, 1999; Crompton, 2000; Kim et al., 2003). Whereas ATP depletion is associated with necrosis, the availability of ATP is required for the development of apoptosis. Postmortem neuropathological evaluation indicated that hemorrhagic necrosis is present in the brain stem of patients who died of acute organophosphate poisoning (de Reuck et al., 1979). It follows that a potential repertoire of cholinergic receptor-independent cellular events that underlie fatal organophosphate poisoning entails the induction of mitochondrial dysfunction because of the severe cellular insult, followed by bioenergetic failure that leads to necrotic cell death because of ATP depletion.

The present study was undertaken to evaluate this hypothesis. We employed mevinphos (3-[dimethoxyphosphinyloxyl]-2-butenoic acid methyl ester; Mev), a US Environmental Protection Agency Toxicity Category I pesticide to provide the cellular insult. As an organophosphate of the P = O type, Mev acts directly in the brain as an acetylcholinesterase inhibitor (Takahashi et al., 1991). Activation of the muscarinic receptors by the accumulated acetylcholine (Chan et al., 2004; Yen et al., 2004) in the rostral ventrolateral medulla (RVLM), a neural substrate that is intimately related to brain stem death (Chan et al., 2005b), triggers Mev intoxication by inducing mitochondrial dysfunction and bioenergetic failure (Yen et al., 2004, 2005). We chose to use PC12 cell, a noradrenergic clonal cell line derived from rat adrenal pheochromocytoma cells (Greene and Tischler, 1976) instead of targeting the RVLM in laboratory animals for several reasons. In addition to sharing common biochemical and physiological characteristics with RVLM neurons (see Section 4), the assumed absence in an in vitro system of acetylcholinesterase, the pharmacological target of Mev, provided the most opportune experimental condition for the evaluation of our hypothesis. Studies using PC12 cells also provide information on the sole participation of neurons without the concern for possible contributions from glial cells or blood vessels that are abundantly present in the brain (Heneka et al., 1998). Based on systematic dissection of the cellular events in PC12 cells, we demonstrated that Mev induced cholinergic receptor-independent necrotic cell death by depressing the activity of Complexes I to IV in the mitochondrial respiratory chain, eliciting reduction in mitochondrial transmembrane potential ($\Delta\Psi_m$), depleting intracellular ATP contents and damaging cell membrane integrity.

2. Methods

2.1. Cell culture

Naïve PC12 cells were purchased from the Culture Collection and Research Center, Food Industry Research and Development Institute (Hsinchu, Taiwan). They were cultured in the absence of nerve growth factor in Dulbecco's modified Eagle's medium (DMEM; Invitrogen Corporation, Grand Island, NY, USA) that was supplemented with 10% horse serum and 5% fetal brain serum (FBS; Invitrogen), and were maintained at 37 °C in a 5% CO_2-humidified incubator (Huang et al., 2002). The medium was changed every 2–3 days. To avoid the effect of cell density on protein synthesis (Garrels and Schubert, 1979), cells were seeded at a density of 2×10^6 cells per poly-L-lysine-coated 150-mm-diameter polystyrene dish.

2.2. Pharmacological agents

Pharmacological agents used in this study included Mev (kindly provided by Sinon Corporation, Taichung County, Taiwan), atropine (Sigma-Aldrich, St. Louis, MO, USA), mecamylamine (RBI, Natick, MA, USA), and a water-soluble formulation of coenzyme Q10 (CoQ10) prepared according to US patent 6,045,826 using polyoxyethanyl 600 cholesteryl sebacate (PCS-600) as the carrier (kindly provided by Dr. Marianna Sikorska, National Research Council, Ottawa, Canada).

2.3. Proteomic analysis

The proteomic map for naïve PC12 cells, or RVLM, liver or lung obtained from adult Sprague–Dawley rats that were anesthetized with intraperitoneal injection of pentobarbital sodium (50 mg/kg), was generated as detailed previously (Huang et al., 2002; Li et al., 2005b). The experimental procedures on laboratory animals were carried out in compliance with the guidelines of our institutional animal care committee; and were according to the National Institutes of Health guide for the care and use of laboratory animals.

Each silver-stained 2-dimensional electrophoresis gel was scanned by an ImageScanner (Amersham Pharmacia Biotech, Uppsala, Sweden), and protein spots were quantified and numbered using the ImageMaster 2D Elite software (Amersham Pharmacia Biotech). For the purpose of this study, protein spots in the domain of pI: 4.5–5.5 and Mr: 20.1–30 kDa were excised from corresponding Coomassie brilliant blue-stained gels, subject to in-gel trypsin digestion, and analyzed by matrix assisted laser desorption ionization time-of-flight (MALDI-TOF) mass spectrometry (Voyager-DE PRO, Applied Biosystems; Foster City, CA, USA). To identify proteins, the measured mono-isotopic masses of peptide were routinely analyzed using MS-Fit (Protein Prospector, UCSF Mass Spectrometry Facility, San Francisco, CA, USA) and MASCOT (Matrix Science, Boston, MA) search programs (Huang et al., 2002; Li et al., 2005b).

2.4. Mev treatment

Seeded PC12 cells were gently removed from the polystyrene dish and suspended in fresh DMEM supplemented with 10% horse serum and 5% FBS. Mev was then added, and cells were incubated with this organophosphate poison for 1 or 3 h at 37 °C in a 5% CO_2-humidified incubator. The cell

suspension was centrifuged, and PC12 cells harvested from the pellet were subject to subsequent experiments after being re-suspended in fresh culture medium. PC12 cells incubated without Mev for the same duration served as vehicle controls; and seeded PC12 cells removed from the polystyrene dish without further experimental manipulation served as the baseline control. In some experiments, CoQ10 or PCS-600 was given together with Mev to establish a causal relationship between the Mev-induced dysfunction of mitochondrial respiratory chain enzyme, depletion of intracellular ATP content and necrotic cell death.

2.5. Cell viability assays

Two assays were used to assess the viability of PC12 cells. In the trypan blue exclusion assay (Biological Industries, Kibbutz Beit, Israel), the number of bright, viable cells and blue-stained, dead cells was counted under a Zeiss microscope (Oberkochen, Germany), aided by a hemocytometer. In the 4-[3-(4-iodophenyl)-2-(4-nitrophenyl)-2H-5-tetrazolio]-1,3-benzene disulfonate (WST-1) reduction assay (Roche Diagnostics GmbH, Mannheim, Germany), the concentration of the water-soluble formazan product, which correlates with the number of metabolically active cells in the suspension, was determined spectrophotometrically (Anthos Labtec Instrument GmbH, Salzburg, Austria) at an absorbance wavelength of 450–690 nm.

2.6. Assays for mitochondrial respiratory enzyme activities

A mitochondria/cytosol fraction kit (BioVision, Mountain View, CA, USA) was used to isolate the mitochondrial fraction of PC12 cells, which was carried out at 4 °C and completed within 2 h after sample collection. The amount of total protein in the mitochondrial suspension was estimated by a protein assay kit (Bio-Rad, Hercules, CA, USA). All assays for the activity of mitochondrial respiratory enzymes, which took place immediately thereafter, were performed using a thermostatically regulated (37 °C) UV/visible spectrophotometer (Amersham Pharmacia Biotech) with reagents purchased from Sigma-Aldrich. Measurement of nicotinamide adenine dinucleotide (NADH) cytochrome *c* reductase (NCCR; enzyme marker for Complexes I and III), succinate cytochrome *c* reductase (SCCR; enzyme marker for Complexes II and III) or cytochrome *c* oxidase (CCO; enzyme marker for Complex IV) activity was performed as reported previously (Chuang et al., 2002; Yen et al., 2004). At least quadruplicate determination was carried out for each sample in all enzyme assays.

2.7. Cytofluorimetric analysis of mitochondrial transmembrane potential

We used 5,5′,6,6′-tetrachloro-1,1′,3,3′-tetraethylbenzimidazolcarbocyanine iodide (JC-1; Molecular Probes, Eugene, OR, USA) to assess $\Delta\Psi_m$ (Cossarizza et al., 1993). Changes in the intensity of red fluorescence associated with J-aggregate formation in mitochondria indicative of maintained $\Delta\Psi_m$ and green fluorescence associated with JC-1 monomers that represent a reduction in $\Delta\Psi_m$ were analyzed by flow cytometry (Beckman Coulter, Fullerton, CA, USA).

2.8. Measurement of ATP concentration, ADP/ATP ratio or lactate dehydrogenase activity

The ATP concentration or ADP/ATP ratio in PC12 cells was measured by an ATP bioluminescence assay (Roche) or an ADP/ATP ratio assay (BioVision). Light emitted from luciferase-mediated reaction and measured by a tube luminometer (Berthold Detection Systems GmbH, Pforzheim, Germany) was used to calculate the respective values. Lactate dehydrogenase (LDH) activity was measured by a LDH-cytotoxicity assay kit (BioVision). The concentration of the formazan product in the medium, which represents LDH activity, was determined spectrophotometrically at an absorbance wavelength of 490–500 nm.

2.9. Flow cytometric or confocal microscopic analysis of necrotic or apoptotic cell death

We determined the nature of cell death using propidium iodide (PI) and fluorescein isothiocyanate-conjugated annexin V (Roche), in conjunction with flow cytometry (Beckman Coulter) or laser scanning confocal microscopy (Fluorview FV300, Olympus, Tokyo, Japan) using differential interference contrast optics and fluorescence detection. Cells that manifested positive stain with PI were taken to represent necrotic cells; and those that exhibited annexin V stain were taken as apoptotic cells (Huang et al., 2002).

2.10. Statistical analysis

All values are expressed as mean ± S.E.M. Student's *t*-test or one-way analysis of variance was used to assess group means. The latter was followed by the Scheffé multiple-range test for post hoc assessment of individual means. $P < 0.05$ was taken to be statistically significant.

3. Results

3.1. Exclusive presence of a neuronal marker in naïve PC12 cells and brain tissues

We used undifferentiated PC12 cells in the present study to avoid the confounding influence of nerve growth factor on cell death (Frederickson et al., 1992). However, to qualify as neurons, it is essential that naïve PC12 cells express the hallmark proteins present in cells of neuronal lineage. For this purpose, we generated a proteomic map for the undifferentiated PC12 cells and compared it against those for the RVLM, liver or lung in the rat. Analysis of the domain of pI: 4.5–5.5 and Mr: 20.1–30 kDa in the silver-stained 2-dimensional electrophoresis gels revealed the omnipresence of three prominent protein spots (Fig. 1A) in naïve PC12 cells and RVLM in rats. Based on complementary search results on samples from 6 different plates of PC12 cells or 6 different animals using MS-Fit and MASCOT programs, which detect matched peptides in the tryptic peptide spectrum with observed m/z values and tryptic cutting sites, we identified spot 1 (Fig. 1B) to be a specific neuronal marker (Wilkinson et al., 1989), ubiquitin carboxyl-terminal hydrolase isozyme L1 (SWISS Prot accession No Q00981). Spots 2 (Fig. 1C) and 3 (Fig. 1D) were identified respectively as peroxiredoxin 2 (SWISS Prot accession No P35704) and phosphatidylethanolamine-binding protein (SWISS Prot accession No P31044). Intriguingly, whereas spots 2 and 3 were present in the proteomic map of PC12 cells and all rat tissues examined, spot 1 is only present in PC12 cells and RVLM, and is conspicuously missing from liver or lung (Fig. 1A).

3.2. Mev reduces PC12 cell viability

Trypan blue exclusion assay (Fig. 2) revealed that at concentrations ranging from 0.2 to 8 μmol, Mev induced a dose-related and time-dependent reduction in the viability of PC12 cells. Whereas WST-1 reduction assay revealed similar trends (Fig. 2), the dose-response curve obtained 1 or 3 h after incubation with Mev exhibited a discernibly steeper slope. Since the WST-1 assay measures mitochondrial dehydrogenase

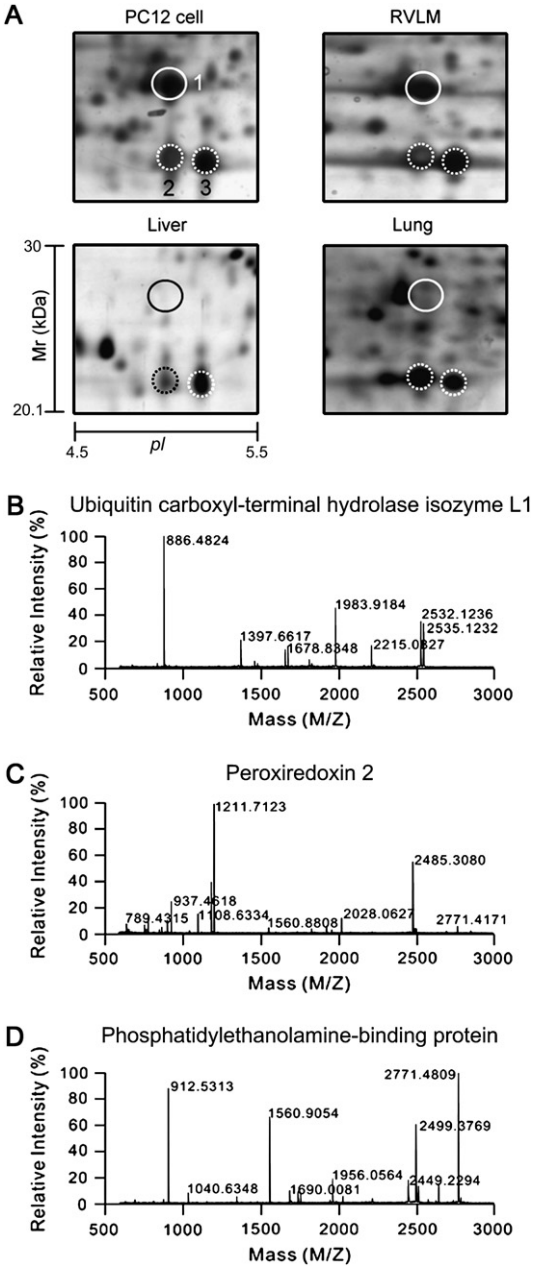

Fig. 1. Representative silver-stained 2-dimensional electrophoresis gel within the domain of pI: 4.5–5.5 and Mr: 20.1–30 kDa of PC12 cells or rostral ventrolateral medulla (RVLM), liver or lung from Sprague–Dawley rat (A), showing the location of protein spot for ubiquitin carboxyl-terminal hydrogenase isozyme L1 (spot 1), peroxiredoxin 2 (spot 2) or phosphatidylethanolamine-binding protein (spot 3). These protein spots were identified by computer matching based on MS-Fit and MASCOT search programs of individual tryptic peptide spectrum (B to D) generated by MALDI-TOF mass spectrometry. Results are representative of six individual samples obtained under basal conditions. Note that the neuronal marker, ubiquitin carboxyl-terminal hydrogenase isozyme L1 is only present in the proteomic map of PC12 cells and RVLM.

activity as an index of cell survival (Huhtala et al., 2002), the manifested response pattern strongly implies that mitochondrial dysfunction underlies the Mev-elicited PC12 cell death. This implication was evaluated in subsequent experiments using two doses of Mev (0.4 or 4 μmol) at two treatment time-points (1 or 3 h). Of note is that after incubation with Mev at 4 μmol for 3 h, the number of viable PC12 cells amounted to only 10% of baseline control.

3.3. Mev-induced PC12 cell death is not mediated via muscarinic or nicotinic receptors

As an inhibitor of acetylcholinesterase (Tafuri and Roberts, 1987; Takahashi et al., 1991), Mev elicits its pharmacological effects on cholinergic receptors via the accumulated acetylcholine. The assumed absence of acetylcholinesterase in our in vitro system therefore provided an ideal experimental condition for the evaluation of cholinergic receptor-independent cellular actions of Mev. Fig. 3 shows that this assumption was justified. Both trypan blue exclusion assay and WST-1 reduction assay indicated that co-incubation with 100 nmol of the muscarinic receptor antagonist (Rabe et al., 1987), atropine or a nicotinic receptor antagonist (Reuben et al., 1998), mecamylamine did not significantly affect the reduction in cell viability induced by Mev.

3.4. Mev depresses mitochondrial respiratory enzyme activities in PC12 cells

The first indication that Mev induces mitochondrial dysfunction in PC12 cells was depression of mitochondrial respiratory enzyme activities (Fig. 4). The activity of NCCR, SCCR or CCO was dose-relatedly and time-dependently reduced after Mev treatment (0.4 or 4 μmol). The three respiratory chain enzymes, however, manifested differential degrees of induced depression. Being the most significantly affected, NCCR activity was already reduced by 50% at 1 h, followed by further reduction (to 65%) 3 h after Mev treatment. Whereas the suppression of activity was the least for SCCR (20–25%), CCO was reduced by >50% 3 h after PC12 cells were incubated with Mev.

3.5. Mev reduces mitochondrial transmembrane potential in PC12 cells

The second indication that Mev induces mitochondrial dysfunction in PC12 cells was a reduction in $\Delta\Psi_m$. As shown in Fig. 5, Mev treatment (0.4 or 4 μmol) resulted in a progressive and dose-dependent increase in JC-1 monomers, which is indicative of a reduction in $\Delta\Psi_m$ (Cossarizza et al., 1993), alongside a corresponding decrease in J-aggregates in PC12 cells. These cellular events were most prominent 3 h after Mev treatment at 4 μmol.

3.6. Mev decreases ATP concentration and increases ADP/ATP ratio in PC12 cells

The third indication that Mev induces mitochondrial dysfunction in PC12 cells was bioenergetic failure (Fig. 6). Whereas ATP concentration was significantly decreased, ADP/ATP ratio was increased in a dose-related and time-dependent manner after treatment with Mev (0.4 or 4 μmol). The intracellular ATP content was also reduced by 75% after incubation of PC12 cells with Mev at 4 μmol for 3 h, paralleled by a 3-fold increase in ADP/ATP ratio.

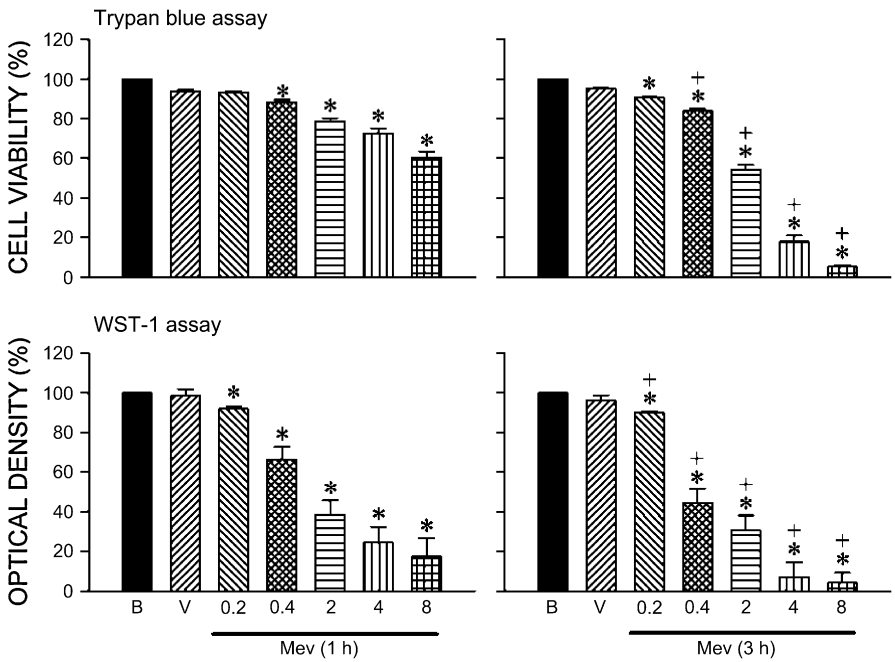

Fig. 2. Effects of Mev (0.2 to 8 μmol) or culture medium (V) on viability of PC12 cells based on trypan blue exclusion assay (upper panels) or WST-1 assay (lower panels). Values are mean ± S.E.M. from 5–7 experiments per treatment group. $^{*}P < 0.05$ versus baseline- (B) or vehicle-control (V) group, and $^{+}P < 0.05$ versus corresponding 1-h treatment group in the Scheffé multiple-range analysis.

3.7. Mev induces LDH release from PC12 cells

Mitochondrial dysfunction may lead to either necrotic or apoptotic cell death (Ankarcrona et al., 1995; Kroemer, 1999; Crompton, 2000; Kim et al., 2003). The first indication that Mev induces necrotic cell death in PC12 cells was damage of plasma membrane. Fig. 6 showed that Mev (0.4 or 4 μmol) dose-relatedly and time-dependently increased LDH activity

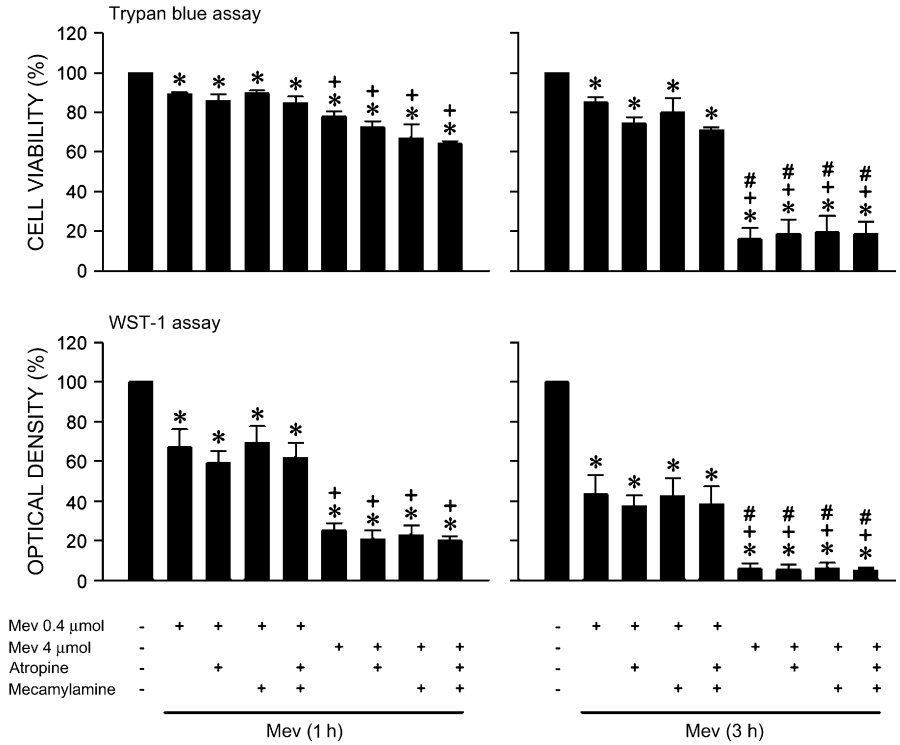

Fig. 3. Effects of co-incubation of atropine (100 nmol) or mecamylamine (100 nmol) on the actions of Mev (0.4 or 4 μmol) on viability of PC12 cells. Values are mean ± S.E.M. from 5–7 experiments per treatment group. $^{*}P < 0.05$ versus baseline-control group, $^{\#}P < 0.05$ versus corresponding 0.4 μmol treatment group, and $^{+}P < 0.05$ versus corresponding 1-h treatment group in the Scheffé multiple range-analysis.

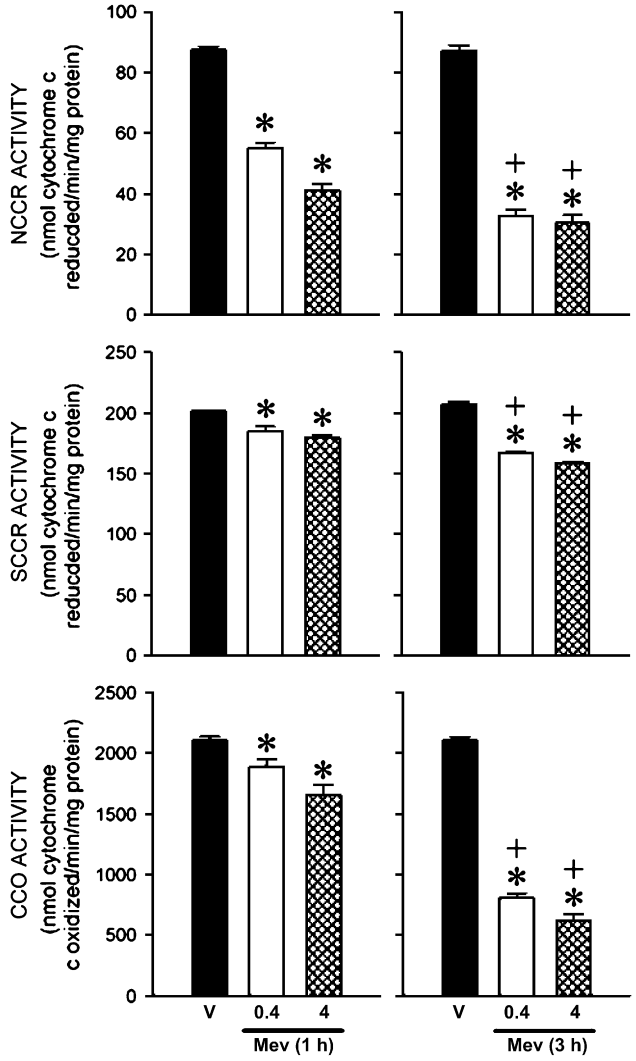

Fig. 4. Effects of Mev (0.4 or 4 μmol) or culture medium (V) on the activity of NADH cytochrome *c* reductase (NCCR), succinate cytochrome *c* reductase (SCCR) or cytochrome *c* oxidase (CCO) in the mitochondria of PC12 cells. Values are mean ± S.E.M. of quadruplicate determinations from 5–7 experiments per treatment group. $^{*}P < 0.05$ versus vehicle-control (V) group, and $^{+}P < 0.05$ versus corresponding 1-h treatment group in the Scheffé multiple-range analysis.

in the medium, an experimental marker of necrotic cell death (Bonfoco et al., 1995). There was a 3-fold increase in LDH activity 3 h after incubation with Mev at 4 μmol.

3.8. Mev induces necrosis in PC12 cells

Flow cytometry confirmed that Mev-induced necrotic cell death indeed took place in PC12 cells. Fig. 7A–C showed that Mev (0.4 or 4 μmol) induced a dose-related and time-dependent surge in the proportion of PC12 cells that stained positively with PI. Quantitative analysis revealed a progressive increase in necrotic PC12 cells (Fig. 7D), along with a minimal presence of annexin V-labeled Mev-treated cells (Fig. 7E) despite positive labeling in PC12 cells subject to serum deprivation (E1). Laser confocal microscopic analysis confirmed that whereas the nucleus of PC12 cells was labeled with PI after Mev treatment (G1,G2), PI-stain is absent in vehicle-control group (F1,F2). On the other hand, annexin V is absent from PC12 cells in either vehicle-control (H1,H2) or Mev-treated (I1,I2) group.

3.9. CoQ10 antagonizes the effects of Mev on mitochondrial respiratory enzyme activities or ATP concentration in, or LDH release from PC12 cells

CoQ10 is a highly mobile electron carrier in the mitochondrial respiratory chain (Ernster and Dallner, 1995) that also acts as an antioxidant (Frei et al., 1990). We demonstrated previously (Li et al., 2005a; Yen et al., 2005) that CoQ10 confers cardiovascular protection against acute Mev intoxication by ameliorating the induced dysfunction of mitochondrial respiratory enzymes or reduced ATP level in the RVLM. CoQ10 was therefore used to establish a causal relationship between the Mev-induced depression of mitochondrial respiratory chain enzyme activities, depletion of intracellular ATP content and necrotic cell death. Table 1 shows that contrast to vehicle (PCS-600), co-incubation of CoQ10 (400 μg/ml) with Mev (4 μmol) for 3 h significantly antagonized the depression of NCCR, SCCR or CCO activity and reduction in ATP concentration elicited by the organophosphate poison in PC12 cells. Intriguingly, CoQ10 also significantly blunted the Mev-induced augmentation of LDH activity in the medium, an experimental index for necrotic cell death.

4. Discussion

The present study took advantage of an experimental strategy that allowed us to investigate the cholinergic receptor-independent cellular mechanisms that underlie fatal organophosphate poisoning. Based on results from PC12 cells, we provided the novel demonstration that by depressing the activity of mitochondrial respiratory chain enzymes, eliciting a reduction in $\Delta\Psi_m$, depleting intracellular ATP content and damaging cell membrane integrity, Mev induces necrotic cell death that is independent of muscarinic or nicotinic receptors.

The PC12 cells share several common biochemical and physiological characteristics with RVLM neurons, whose activity reflects the "life-and-death" status in patients (Chan et al., 2005b). Both PC12 cells and RVLM neurons are catecholamine-containing cells (Greene and Tischler, 1976; Ross et al., 1984), and both respond to *Escherichia coli* lipopolysaccharide or Mev by producing nitric oxide via activation of the inducible nitric oxide synthase (Heneka et al., 1998; Chan et al., 2001, 2004, 2005a; Li et al., 2005b). The present study further demonstrated that undifferentiated PC12 cells expressed the specific neuronal marker, ubiquitin carboxyl-terminal hydrolase isozyme L1, a reduction in the RVLM of which exacerbates Mev-induced fatality in rats (Chang et al., 2004) or lipopolysaccharide-induced toxicity in PC12 cells (Chang and Chan, unpublished observations). Thus, it is

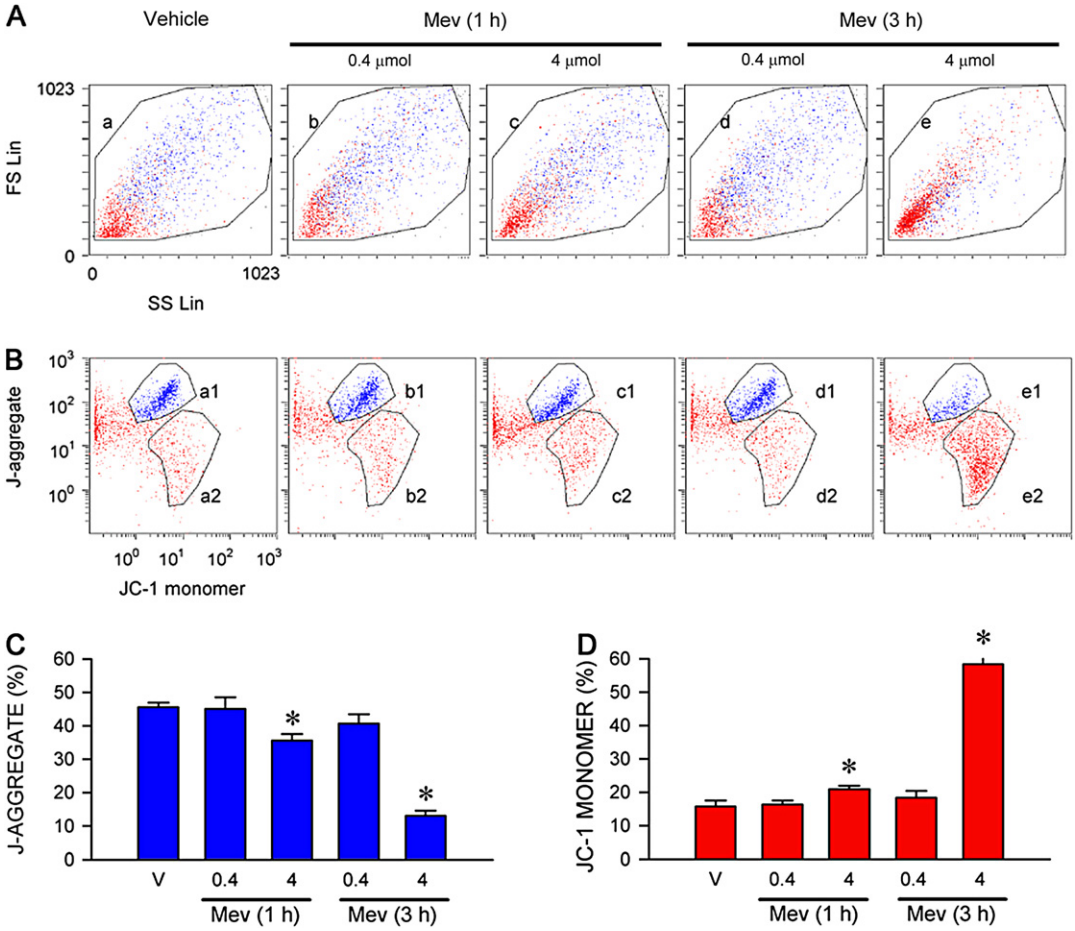

Fig. 5. Representative flow cytometric profiles showing changes in light scatter dot plot (A) or subpopulations (B) of J-aggregates (a1–e1) and JC-1 monomers (a2–e2) in PC12 cells treated with Mev (0.4 or 4 μmol) or culture medium (V). The proportion of PC12 cells that showed J-aggregates that indicates maintained $\Delta\Psi_m$ (C) or JC-1 monomers that represents reduced $\Delta\Psi_m$ (D) is presented in mean ± S.E.M. from 5–7 experiments per treatment group. $^*P < 0.05$ versus vehicle-control (V) group in the Scheffé multiple-range analysis.

intriguing that, based on this clinically relevant in vitro model, the present study demonstrated that the repertoire of cholinergic receptor-independent cellular events that underlie fatal organophosphate poisoning entails mitochondrial dysfunction, followed by bioenergetic failure that leads to necrotic cell death because of ATP depletion.

During oxidative phosphorylation (Hatefi, 1985), electrons from the reducing agent, NADH, move from Complex I through coenzyme Q to Complex III, and then through cytochrome *c* to Complex IV. Electrons from succinate enter the respiratory chain through flavin adenine dinucleotide (FAD), which is covalently linked to Complex II, and move to Complexes III and IV by way of CoQ10. The passage of electrons between these complexes releases energy that is stored in the form of a proton gradient across the inner membrane, and is used by F_1F_0-ATPase to produce ATP from ADP. Results from the present study indicated that the activity of NCCR, SCCR and CCO in PC12 cells was depressed after prolonged Mev treatment. It follows that our observed Mev-induced cell death may arise from a dysfunction of Complexes I through IV in the mitochondrial electron transport chain. It is also likely that biochemical pathways linking to NADH and FAD are engaged in the observed progressive reduction in intracellular ATP contents.

We previously found that under conditions of Mev intoxication that are not fatal, this organophosphate poison depresses only NCCR and CCO activity in the RVLM of rats without affecting SCCR activity (Yen et al., 2004). On the other hand, the activity of all three respiratory chain enzymes in the RVLM was significantly reduced on administration of a lethal dose of Mev (Li et al., 2005a). The implied notion that SCCR activity is a crucial determinant for fatality is in line with the stipulation that the ratio between glycolytic and respiratory ATP generation is proportional to apoptosis/necrosis susceptibility (Kroemer et al., 1999). Since the primary source of metabolic energy in neurons is respiratory ATP generation, neurons tend to undergo necrosis in response to stress (Kroemer et al., 1999). Aerobic metabolism of glucose is coupled to oxidative phosphorylation by linking to SCCR, FAD and Complex II through succinate in the citric acid cycle. Interestingly, our results showed that whereas the activity of NCCR and CCO in

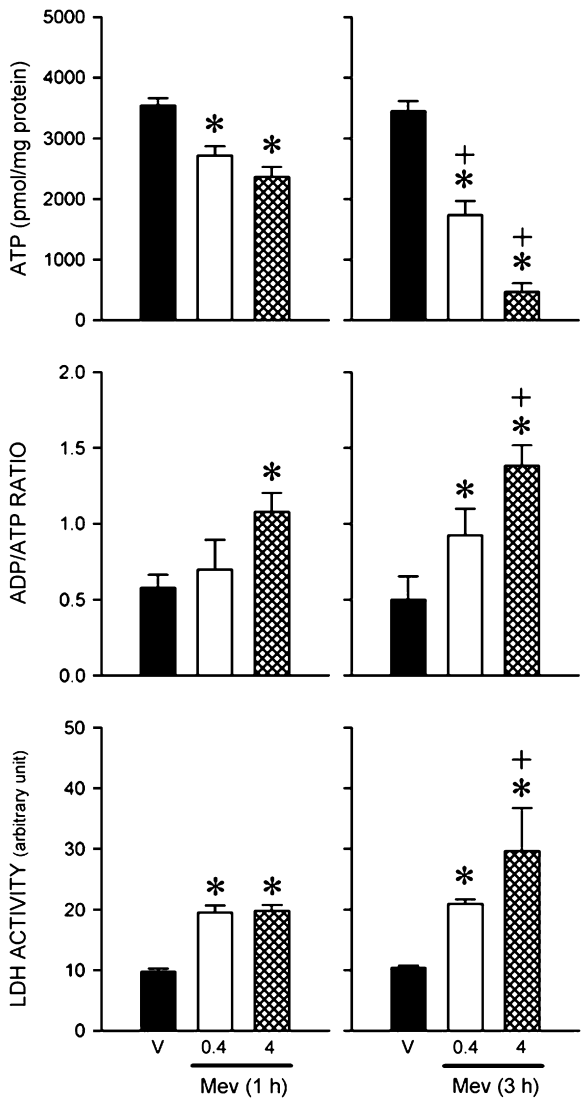

Fig. 6. Effects of Mev (0.4 or 4 μmol) or culture medium (V) on intracellular ATP content (upper panels) or ADP/ATP ratio (middle panels) in PC12 cells, or LDH activity in the medium (lower panels). Values are mean ± S.E.M. from 5–7 experiments per treatment group. $^{*}P < 0.05$ versus vehicle-control (V) group and $^{+}P < 0.05$ versus corresponding 1-h treatment group in the Scheffé multiple-range analysis.

PC12 cells was depressed by more than 50% 3 h after treatment with Mev, the activity of SCCR was reduced by only 20–25%. Nonetheless, such a degree of insufficiency in SCCR activity may already result in reduced ATP production in PC12 cells after treatment with Mev, leading to our observed decrease in intracellular ATP content and increase in ADP/ATP ratio, followed by necrotic cell death. This notion was further strengthened by our results with CoQ10 treatment, which established a causal relationship between the Mev-induced depression of NCCR, SCCR or CCO activity, depletion of intracellular ATP content and necrotic cell death.

Cellular insults may elicit a generalized increase in permeability of the inner mitochondrial membrane, possibly due to breakdown of the phospholipid bilayer (Gunter and Pfeiffer, 1990). It is conceivable that our observed Mev-elicited decrease in $\Delta\Psi_m$ is related to activation of the speculated mitochondrial permeability transition, leading to collapse of the proton-motive force, disruption of ionic homeostasis, loss of pyridine nucleotides and hydrolysis of ATP (Fontaine and Bernardi, 1999). The very last cellular event provides another account for the 75% decrease in ATP content and 3-fold increase in ADP/ATP ratio seen after treating PC12 cells with Mev. It may also be responsible for the rupture of plasma membrane (Kim et al., 2003), as demonstrated by the 3-fold increase in LDH activity in the culture medium. Furthermore, as a major parameter that regulates the open-close transition of permeability pores, the $\Delta\Psi_m$ is suggested to be a key element in the control of proton pumping activity in Complex I (Fontaine and Bernardi, 1999). Thus, our observed reduction in $\Delta\Psi_m$ may also be related to Mev-induced depression of this respiratory complex.

We are aware that acetylcholine is synthesized, stored and released by PC12 cells (Green and Rein, 1977). Furthermore, some authors demonstrated that these clonal cells express acetylcholinesterase although significant augmentation of enzyme activity requires the presence of nerve growth factor (Lucas et al., 1980; Das and Barone, 1999). These reports, however, do not contradict our observed cholinergic receptor-independent effects of Mev because naïve PC12 cells cultured in the absence of nerve growth factor were used in this study; and the dose-related and time-dependent loss of cell viability induced by this organophosphate poison was not reversed by muscarinic (atropine) or nicotinic (mecamylamine) blockade. The organophosphate compound soman increases dopamine and serotonin in hypothalamic tissues that is not related to acetylcholine accumulation (Coudray-Lucas et al., 1987). The possibility therefore exists for these monoamines to be involved in Mev neurotoxicity.

In conclusion, based on an in vitro model using naïve PC12 cells, the present study demonstrated that the repertoire of cellular events that underlie fatal intoxication induced by Mev may entail depression of the activity of Complexes I to IV in the mitochondrial respiratory chain, reduction in $\Delta\Psi_m$, depletion of intracellular ATP content and damage of cell membrane integrity, leading eventually to necrotic cell death. Critically ill patients who succumbed to organophosphate poisoning (Yen et al., 2000) manifest before death a dramatic reduction or loss of a "life-and-death" signal that our laboratory identified from systemic arterial pressure signals (Chan et al., 2005b). This signal is related to the functional integrity of the brain stem (Kuo et al., 1997b) and takes origin from the RVLM (Kuo et al., 1997a). Thus, our demonstration that Mev may elicit fatality by inducing necrotic cell death via a cholinergic receptor-independent action in P12 cells, which share common biochemical and physiological characteristics with RVLM neurons, is of clinical significance. This information opens a new vista in our search for therapeutic remedies against fatal organophosphate poisoning and management strategies against brain stem death.

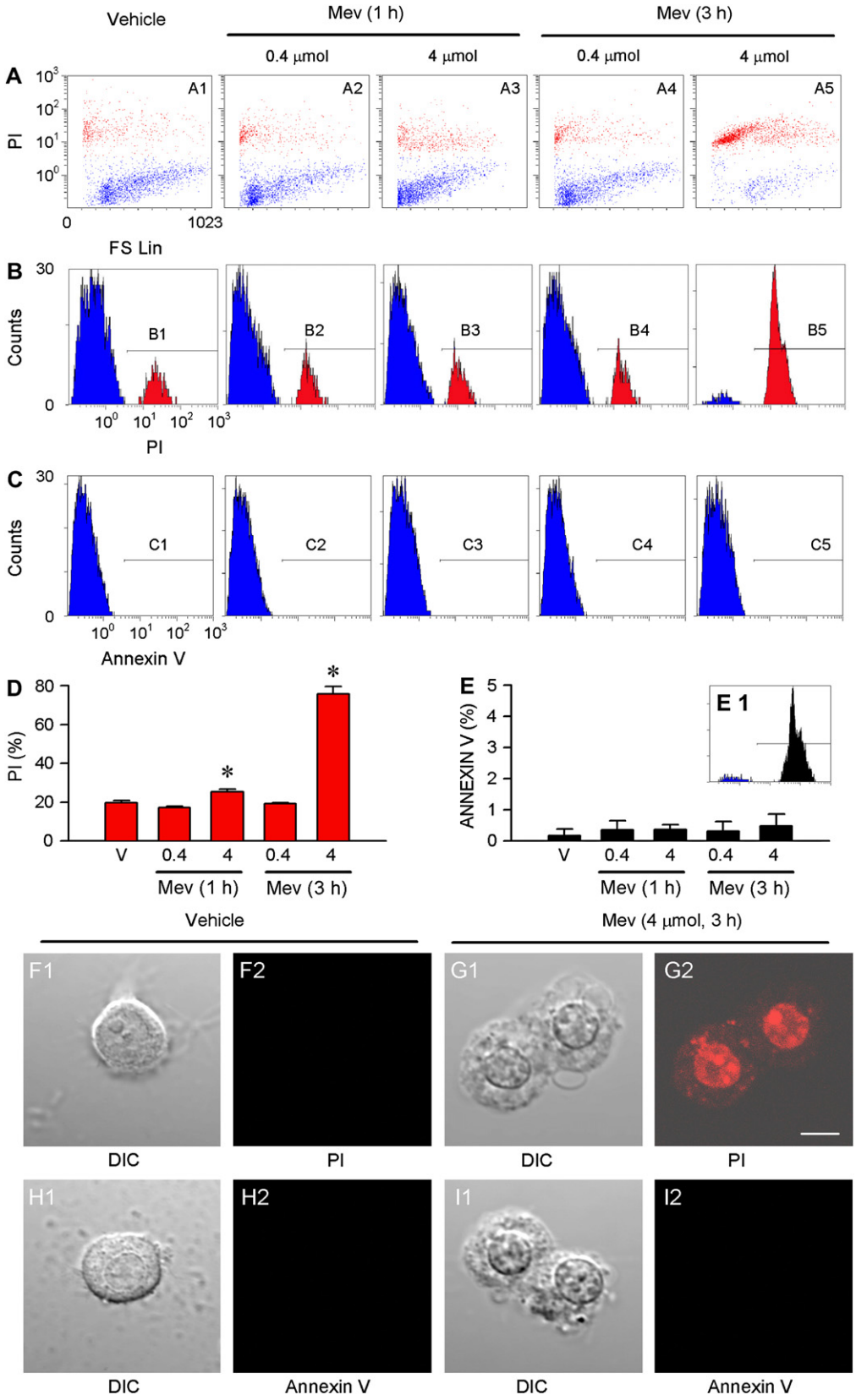

Fig. 7. Representative flow cytometric profiles showing changes in light scatter dot plot (A) or quantification of PC12 cells labeled with propidium iodide (PI; B1−B5) or stained with annexin V (C1−C5) after incubation with Mev (0.4 or 4 μmol) or culture medium (V). Annexin V-labeling obtained from PC12 cells subject to serum deprivation (E1) is included as a positive control. The proportion of PC12 cells that showed PI (D) or annexin V (E) is presented in mean ± S.E.M. from 5−7 experiments per treatment group. $*P < 0.05$ versus vehicle-control (V) group in the Scheffé multiple-range analysis. Panels F to I are representative images of PC12 cells from vehicle-control (F, H) or Mev-treated (G, I) group taken under differential interference contrast optics (DIC) or by laser confocal microscopy for detection of PI or annexin V fluorescence. Scale bar, 5 μm.

Table 1
Effects of coenzyme Q10 (CoQ10, 400 μg/ml) on the activity of mitochondrial respiratory chain enzymes or intracellular ATP content in PC12 cells, or LDH activity in the medium, 3 h after incubation with Mev (4 μmol)

	Mev + PCS-600	Mev + CoQ10
NCCR activity (nmol cytochrome c reduced/min/mg protein)	29.3 ± 2.6	53.5 ± 3.3*
SCCR activity (nmol cytochrome c reduced/min/mg protein)	156.9 ± 8.8	180.5 ± 7.6*
CCO activity (nmol cytochrome c oxidized/min/mg protein)	623.7 ± 55.3	1658.3 ± 80.2*
ATP content (pmol/mg protein)	466.9 ± 144.6	1629.3 ± 138.6*
LDH activity (arbitrary unit)	31.6 ± 8.2	20.8 ± 8.4*

Values are mean ± S.E.M. from 4–5 experiments per treatment group. *$P < 0.05$ versus Mev + PCS-600 group in the Student's *t*-test.

Acknowledgements

This study was supported by research grants NSC-94-2320-B-110-006 (A.Y.W.C.), and NSC-94-2752-B-110-001-PAE and NSC-94-2752-B-110-002-PAE (A.Y.W.C. and S.H.H.C.) from the National Science Council, Taiwan, Republic of China. S.H.H.C. is National Chair Professor of Neuroscience appointed by the Ministry of Education, and Sun Yat-sen Research Chair Professor appointed by the National Sun Yat-sen University, Taiwan, Republic of China.

References

Ankarcrona, M., Dypbukt, J.M., Bonfoco, E., Zhivotovsky, B., Orrenius, S., Lipton, S.A., Nicotera, P., 1995. Glutamate-induced neuronal death: a succession of necrosis or apoptosis depending on mitochondrial function. Neuron 5, 961–973.

Anonymous, 1981. Guidelines for the determination of death. Report of the medical consultants on the diagnosis of death to the President's Commission for the study of Ethical Problems in Medicine and Biomedical and Behavioral Research. JAMA 246, 2184–2186.

Balali-Mood, M., Shariat, M., 1998. Treatment of organophosphate poisoning. Experience of nerve agents and acute pesticide poisoning on the effects of oximes. Journal of Physiology 92, 375–378.

Bardin, P.G., van Eeden, S.F., 1990. Organophosphate poisoning: grading the severity and comparing treatment between atropine and glycopyrrolate. Critical Care Medicine 18, 956–960.

Bardin, P.G., van Eeden, S.F., Moolman, J.A., Foden, A.P., Joubert, J.R., 1994. Organophosphate and carbamate poisoning. Archives of Internal Medicine 154, 1433–1441.

Bonfoco, E., Krainc, D., Ankarcrona, M., Nicotera, P., Lipton, S.A., 1995. Apoptosis and necrosis: two distinct events induced, respectively, by mild and intense insults with *N*-methyl-D-aspartate or nitric oxide/superoxide in cortical cell cultures. Proceedings of the National Academy of Sciences of the United States of America 92, 7162–7166.

Chan, J.Y.H., Wang, S.H., Chan, S.H.H., 2001. Differential roles of iNOS and nNOS at rostral ventrolateral medulla during experimental endotoxemia in the rat. Shock 15, 65–72.

Chan, J.Y.H., Chan, S.H.H., Chang, A.Y.W., 2004. Differential contributions of NOS isoforms in the rostral ventrolateral medulla to cardiovascular responses associated with mevinphos intoxication in the rat. Neuropharmacology 46, 1184–1194.

Chan, J.Y.H., Chan, S.H.H., Li, F.C.H., Cheng, H.L., Chang, A.Y.W., 2005a. Phasic cardiovascular responses to mevinphos are mediated through differential activation of cGMP/PKG cascade and peroxynitrite via nitric oxide generated in the rat rostral ventrolateral medulla by NOS I and II isoforms. Neuropharmacology 48, 161–172.

Chan, J.Y.H., Chang, A.Y.W., Chan, S.H.H., 2005b. New insights on brain stem death: From bedside to bench. Progress in Neurobiology 77, 396–425.

Chang, C., Chang, A.Y.W., Chan, S.H.H., 2004. De novo synthesis of ubiquitin carboxyl-terminal hydrolase isozyme L1 in rostral ventrolateral medulla is crucial to survival during mevinphos intoxication. Shock 22, 575–581.

Chuang, Y.C., Tsai, J.L., Chang, A.Y.W., Chan, J.Y.H., Liou, C.W., Chan, S.H.H., 2002. Dysfunction of the mitochondrial respiratory chain in the rostral ventrolateral medulla during experimental endotoxemia in the rat. Journal of Biomedical Science 9, 542–548.

Conference of Medical Royal Colleges and their Faculties in the United Kingdom, 1976. Diagnosis of brain death. British Medical Journal ii, 1187–1188.

Conference of Medical Royal Colleges and their Faculties in the United Kingdom, 1979. Diagnosis of brain death. British Medical Journal i, 3320.

Cossarizza, A., Baccarani-Contri, M., Kalashnikova, G., Franceschi, C., 1993. A new method for the cytofluorimetric analysis of mitochondrial membrane potential using the J-aggregate forming lipophilic cation 5,5′,6,6′-tetrachloro-1,1′,3,3′-tetraethylbenzimidazolcarbocyanine iodide (JC-1). Biochemical and Biophysical Research Communications 197, 40–45.

Coudray-Lucas, C., Le Guen, A., Prioux-Guyonneau, M., Cohen, Y., Wepierre, J., 1987. Changes in brain monoamine content and metabolism induced by paraoxon and soman intoxication. Effect of atropine. Xenobiotica 17, 1131–1138.

Crompton, M., 2000. Mitochondrial intermembrane junctional complexes and their role in cell death. Journal of Physiology (London) 529, 11–21.

Das, K.P., Barone Jr., S., 1999. Neuronal differentiation in PC12 cells is inhibited by chlorpyrifos and its metabolites: is acetylcholinesterase inhibition the site of action? Toxicology and Applied Pharmacology 160, 217–230.

de Reuck, J., Colardyn, F., Willems, J., 1979. Fatal encephalopathy in acute poisoning with organophosphorus insecticides: a clinico-pathologic study of two cases. Clinical Neurology and Neurosurgery 81, 247–254.

Eguchi, Y., Shimizu, S., Tsujimoto, Y., 1997. Intracellular ATP levels determine cell death fate by apoptosis or necrosis. Cancer Research 57, 1835–1840.

Ernster, L., Dallner, G., 1995. Biochemical, physiological and medical aspects of ubiquinone function. Biochimica Biophysica Acta 1271, 195–204.

Fontaine, E., Bernardi, P., 1999. Progress on the mitochondrial permeability transition pore: regulation by complex I and ubiquinone analogs. Journal of Bioenergetics and Biomembranes 31, 335–345.

Frederickson, R.M., Mushynski, W.E., Sonenberg, N., 1992. Phosphorylation of translation initiation factor eIF-4E is induced in a ras-dependent manner during nerve growth factor-mediated PC12 cell differentiation. Molecular and Cellular Biology 12, 1239–1247.

Frei, B., Kim, M.C., Ames, B.N., 1990. Ubiquinol-10 is an effective lipid-soluble antioxidant at physiological concentrations. Proceedings of the National Academy of Sciences of the United States of America 87, 4879–4883.

Garrels, J.I., Schubert, D., 1979. Modulation of protein synthesis by nerve growth factor. Journal of Biological Chemistry 254, 7978–7985.

Green, L.A., Rein, G., 1977. Synthesis, storage and release of acetylcholine by a noradrenergic pheochromocytoma cell line. Nature 268, 349–351.

Greene, L.A., Tischler, A.S., 1976. Establishment of a noradrenergic clonal line of rat adrenal pheochromocytoma cells which respond to nerve growth factor. Proceedings of the National Academy of Sciences of the United States of America 73, 2424–2428.

Gunter, T.E., Pfeiffer, D.R., 1990. Mechanisms by which mitochondria transport calcium. American Journal of Physiology 258, C755–C786.

Hatefi, Y., 1985. The mitochondrial electron transport and oxidative phosphorylation system. Annual Review of Biochemistry 54, 1015–1069.

Heneka, M.T., Löschmann, R.-A., Gleichmann, M., Weller, M., Schulz, J.B., Wüllner, U., Klockgether, T., 1998. Induction of nitric oxide synthase and nitric oxide mediated apoptosis in neuronal PC12 cells after stimulation with tumor necrosis factor-α/lipopolysaccharide. Journal of Neurochemistry 71, 88–94.

Huang, Y.H., Chang, A.Y.W., Huang, C.M., Huang, S.W., Chan, S.H.H., 2002. Proteomic analysis of lipopolysaccharide-induced apoptosis in PC12 cells. Proteomics 2, 1220–1228.

Huhtala, A., Mannerstrom, M., Alajuuma, P., Nurmi, S., Toimela, T., Tahti, H., Salminen, L., Uusitalo, H., 2002. Comparison of an immortalized human corneal epithelial cell line and rabbit corneal epithelial cell culture in cytotoxicity testing. Journal of Ocular Pharmacology & Therapeutics 18, 163–175.

Hung, T.P., Chen, S.T., 1995. Prognosis of deeply comatose patients on ventilators. Journal of Neurology. Neurosurgery & Psychiatry 58, 75–80.

Kim, J.S., He, L., Lemasters, J.J., 2003. Mitochondrial permeability transition: a common pathway to necrosis and apoptosis. Biochemical and Biophysical Research Communications 304, 463–470.

Kroemer, G., 1999. Mitochondrial control of apoptosis: an overview. Biochemical Society Symposia 66, 1–15.

Kuo, T.B.J., Yang, C.C.H., Chan, S.H.H., 1997a. Selective activation of vasomotor component of SAP spectrum by nucleus reticularis ventrolateralis in rats. American Journal of Physiology 272, H485–H492.

Kuo, T.B.J., Yien, H.W., Hseu, S.S., Yang, C.C.H., Lin, Y.Y., Lee, L.C., Chan, S.H.H., 1997b. Diminished vasomotor component of systemic arterial pressure signals and baroreflex in brain death. American Journal of Physiology 273, H1291–H1298.

Lallement, G., Baubichon, D., Clarencon, D., Galonnier, M., Peoc'h, M., Carpentier, P., 1999. Review of the value of gacyclidine (GK-11) as adjuvant medication to conventional treatments of organophosphate poisoning: primate experiments mimicking various scenarios of military or terrorist attack by soman. Neurotoxicology 20, 675–684.

Leist, M., Single, B., Castoldi, A.F., Kühnle, S., Nicotera, P., 1997. Intracellular adenosine triphosphate (ATP) concentration: a switch in the decision between apoptosis and necrosis. Journal of Experimental Medicine 185, 1481–1486.

Li, F.C.H., Tseng, H.P., Chang, A.Y.W., 2005a. Neuroprotective role of coenzyme Q10 against dysfunction of mitochondrial respiratory chain at rostral ventrolateral medulla during fatal mevinphos intoxication in the rat. Annals of the New York Academy of Science 1042, 1–8.

Li, F.C.H., Chan, J.Y.H., Chan, S.H.H., Chang, A.Y.W., 2005b. In the rostral ventrolateral medulla, the 70 kDa heat shock protein (HSP70), but not HSP90, confers neuroprotection against fatal endotoxemia via augmentation of nitric oxide synthase I (NOS I)/protein kinase G signaling pathway and inhibition of NOS II/peroxynitrite cascade. Molecular Pharmacology 68, 179–192.

Lucas, C.A., Czlonkowska, A., Kreutzberg, G.W., 1980. Regulation of acetylcholinesterase by nerve growth factor in the pheochromocytoma PC12 cell line. Neuroscience Letters 18, 333–337.

McCauley, L.A., Rischitelli, G., Lambert, W.E., Lasarev, M., Sticker, D.L., Spencer, P.S., 2001. Symptoms of Gulf War veterans possibly exposed to organophosphate chemical warfare agents at Khamisiyah. Iraq International Journal of Occupational and Environmental Health 7, 79–89.

Rabe, C.S., Delorme, E., Weight, F.F., 1987. Muscarine-stimulated neurotransmitter release from PC12 cells. Journal of Pharmacology and Experimental Therapeutics 243, 534–541.

Reuben, M., Louis, M., Clarke, P.B., 1998. Persistent nicotinic blockade by chlorisondamine of noradrenergic neurons in rat brain and cultured PC12 cells. British Journal of Pharmacology 125, 1218–1227.

Ross, C.A., Ruggiero, D.A., Park, D.H., Joh, T.H., Sved, A.F., Fernandez-Pardal, J., Saavedra, J.M., Reis, D.J., 1984. Tonic vasomotor control by the rostral ventrolateral medulla: effect of electrical or chemical stimulation of the area containing C_1 adrenaline neurons on arterial pressure, heart rate, and plasma catecholamines and vasopressin. Journal of Neuroscience 4, 474–494.

Tafuri, J., Roberts, J., 1987. Organophosphate poisoning. Annals of Emergency Medicine 16, 193–202.

Takahashi, H., Kojima, T., Ikeda, T., Tsuda, S., Shirasu, Y., 1991. Differences in the mode of lethality produced through intravenous and oral administration of organophosphorus insecticides in rats. Fundamental and Applied Toxicology 16, 459–468.

Uchiyama, T., Otani, H., Okada, T., Ninomiya, H., Kido, M., Imamura, H., Nogi, S., Kobayashi, Y., 2002. Nitric oxide induces caspase-dependent apoptosis and necrosis in neonatal rat cardiomyocytes. Journal of Molecular and Cellular Cardiology 34, 1049–1061.

Wilkinson, K.D., Lee, K., Deshpande, S., Duerksen-Hughes, P., Boss, J.M., Poel, J., 1989. The neuron-specific protein PGP9.5 is a ubiquitin carboxyl-terminal hydrolase. Science 246, 670–673.

Yen, D.H.T., Yien, H.W., Wang, L.M., Lee, C.H., Chan, S.H.H., 2000. Spectral analysis of systemic arterial pressure and heart rate signals of patients with acute respiratory failure induced by severe organophosphate poisoning. Critical Care Medicine 28, 2805–2811.

Yen, D.H.T., Chan, J.Y.H., Tseng, H.P., Huang, C.L., Lee, C.H., Chan, S.H.H., Chang, A.Y.W., 2004. Depression of mitochondrial respiratory enzyme activity in rostral ventrolateral medulla during acute mevinphos intoxication in the rat. Shock 21, 358–363.

Yen, D.H.T., Chan, J.Y.H., Huang, C.I., Lee, C.H., Chan, S.H.H., Chang, A.Y.W., 2005. Coenzyme Q10 confers cardiovascular protection against acute mevinphos intoxication by ameliorating bioenergetic failure and hypoxia in the rostral ventrolateral medulla of the rat. Shock 23, 353–359.